Mathematical Methods MTH1002 Term 1

Exercise Sheet for Week 9

Key:

- **Questions marked with a ☙** are difficult questions.
- **Questions marked with a ♫** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

♫ **Exercise 1.**

(i) State the Sandwich Theorem for sequences.

(ii) Find

(a) $\lim_{n\to\infty} \dfrac{3n}{n-3\sin n}$

(b) $\lim_{n\to\infty} \left(n+\sqrt{n}\right)^{\frac{1}{n}}$.

♫ **Exercise 2.** Find the limits of the following sequences (if they exist):

(i) $\lim_{n\to\infty} 2^{\frac{1}{n}}$

(ii) $\lim_{n\to\infty} \dfrac{n-4}{2n+3}$

(iii) $\lim_{n\to\infty} \dfrac{3n^2+n+2}{n+2}$

(iv) $\lim_{n\to\infty} (3^n+1)^{\frac{1}{n}}$

(v) $\lim_{n\to\infty} \dfrac{5^n-2^{-n}}{5^{n+2}}$

(vi) $\lim_{n\to\infty} (-1)^n \dfrac{4n^4}{(3n^2+1)(n^2-1)}$

(vii) $\lim_{n\to\infty} \dfrac{\pi^n}{e^n}$

(viii) $\lim_{n\to\infty} \cos\left(\dfrac{\pi}{4}+\dfrac{1}{n}\right)$

(ix) $\lim_{n\to\infty} \sqrt[n]{3^n+n+2}$

♫ **Exercise 3.** We have met the following theorems.

- If an infinite sequence is monotonic increasing and bounded above then it must converge to a finite limit.
- If an infinite sequence is monotonic decreasing and bounded below then it must converge to a finite limit.

For any sequence, determine if the following statements are true or false, and if false, give a counterexample:

(i) If $\{a_n\}$ is bounded, then it converges.

(ii) If $\{a_n\}$ is not bounded, then it diverges.

(iii) If $\{a_n\}$ diverges, then it is not bounded.

Exercise 4.

♫ (i) Suppose that $a \in \mathbb{R}$ where $0 < a < 1$, determine $\lim_{n\to\infty} \dfrac{a^n + a^{-n}}{1 + a^n}$.

(ii) Suppose that $a_{n+1} = \dfrac{-5a_n}{2n+1}$ for $n = 1, 2, \ldots$ and $a_1 = 1$. Find an expression for a_{n+1} in terms of n, and show that $a_{n+1} \to 0$ as $n \to \infty$.